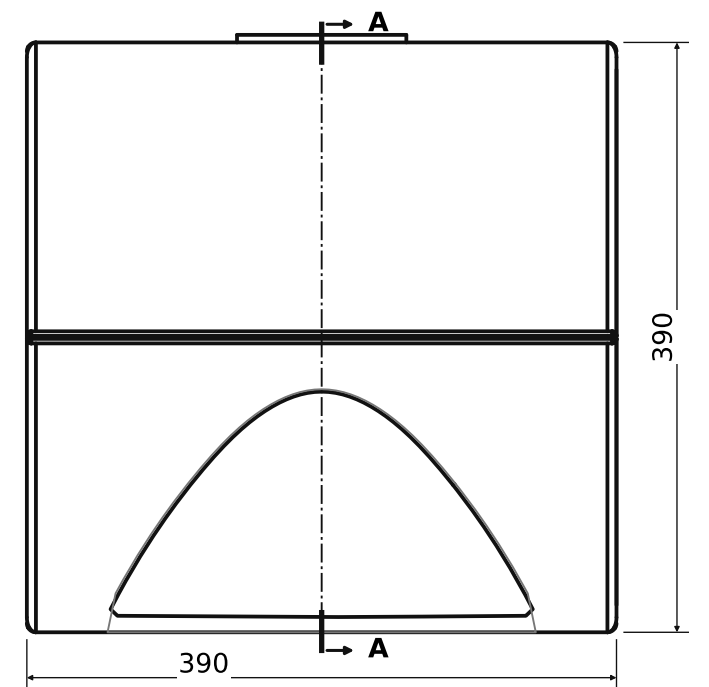
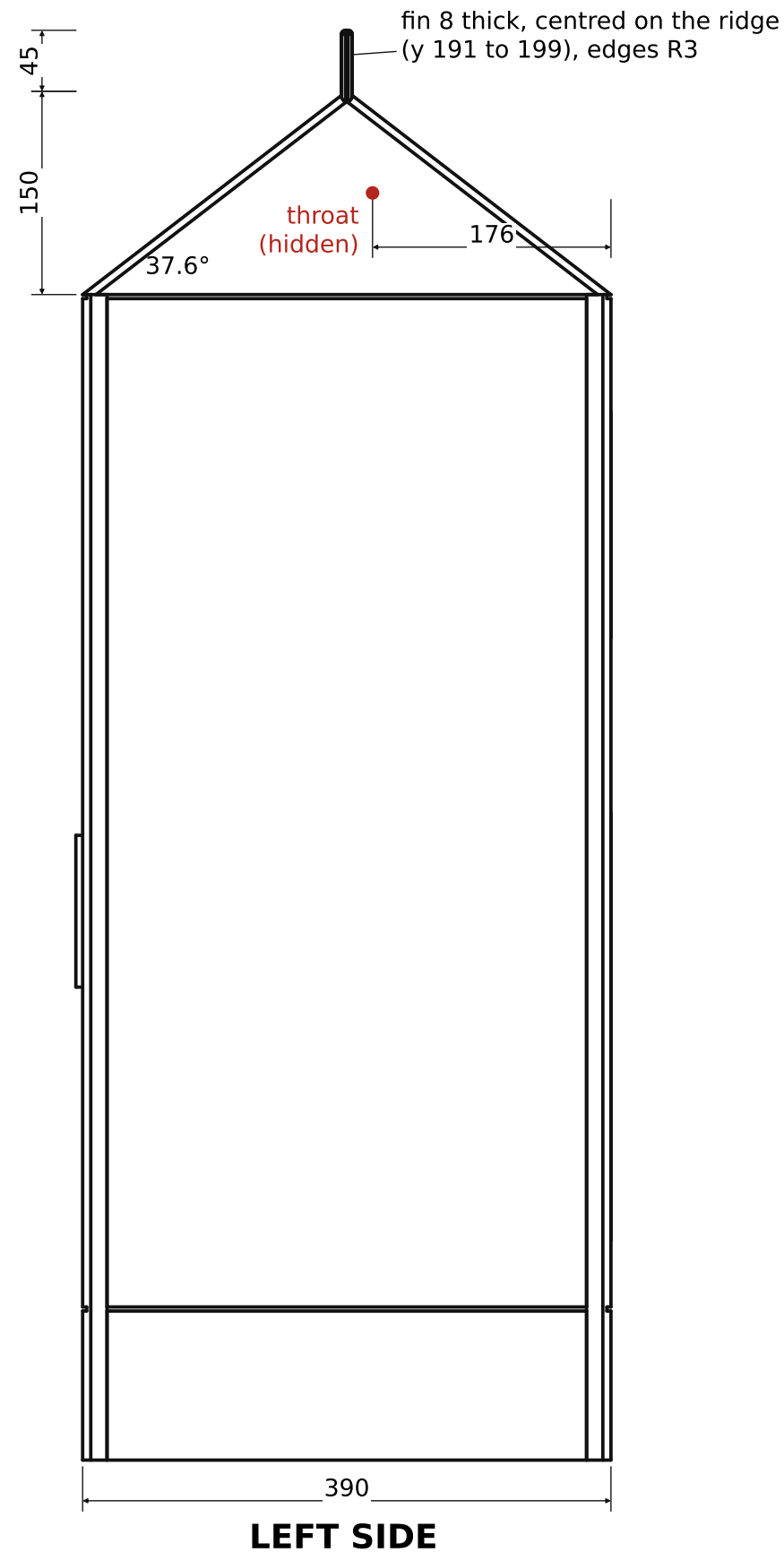
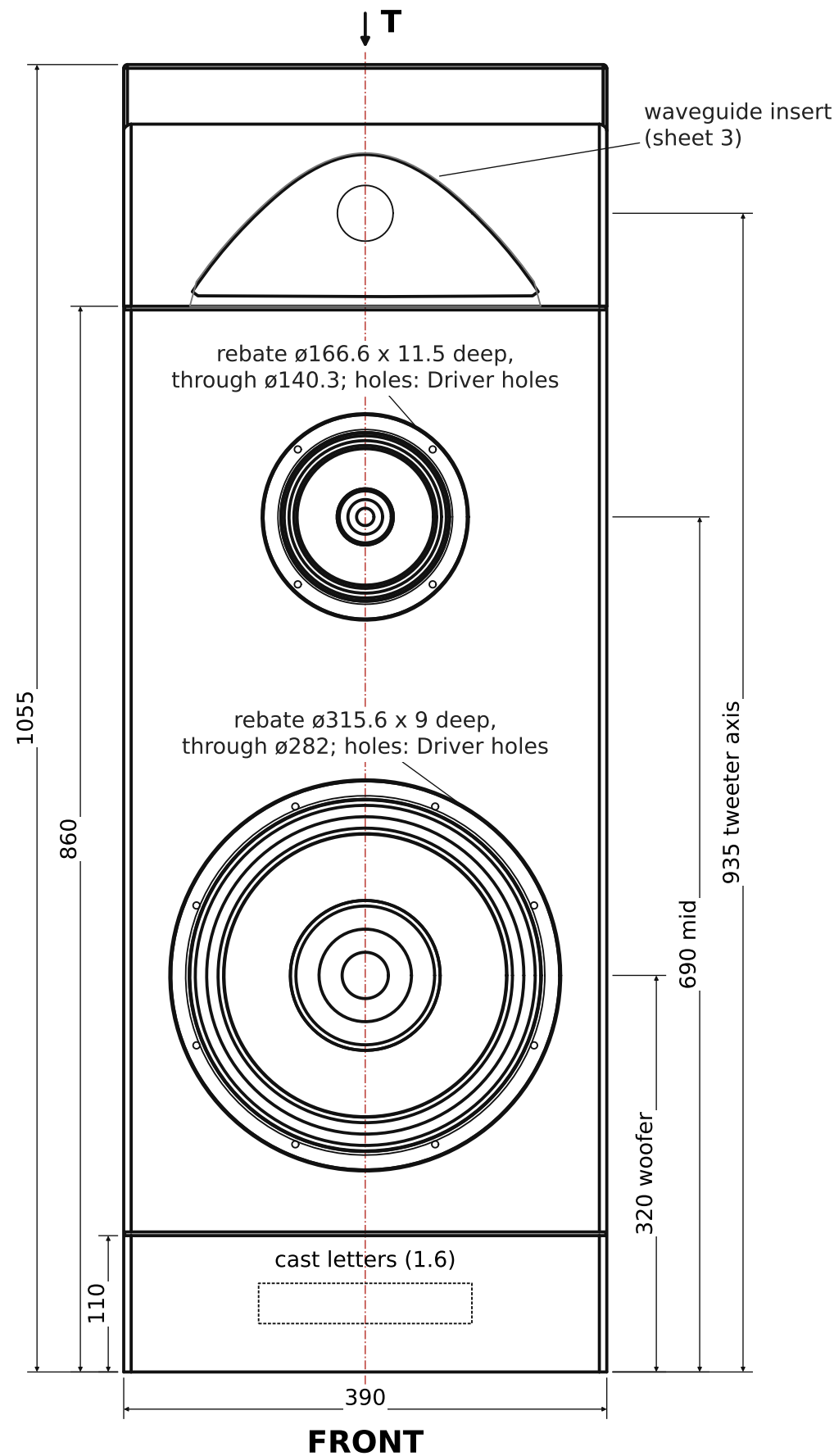




VIEW T (FROM ABOVE)

black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes: notes 1.1 to 1.6, sheet 7

Driver holes (front baffle, from its front face)

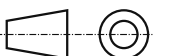
Woofer, centre z 320 on the centreline (M2)
rebate $\varnothing 315.6 \times 9$ deep; through $\varnothing 282$; 8 x $\varnothing 5.5$ on $\varnothing 295$, the first 22.5° above the horizontal, 45° apart, for M4 T-nuts (1.2).

Mid, centre z 690 on the centreline (M3)

rebate $\varnothing 166.6 \times 11.5$ deep; through $\varnothing 140.3$; 4 x $\varnothing 5.5$ on $\varnothing 154$, the first 45° above the horizontal, 90° apart, for M4 T-nuts (1.2).

General arrangement

earmilk floorstander. Front, left side and top at 1:5. Heights from the floor; depths from the front face.

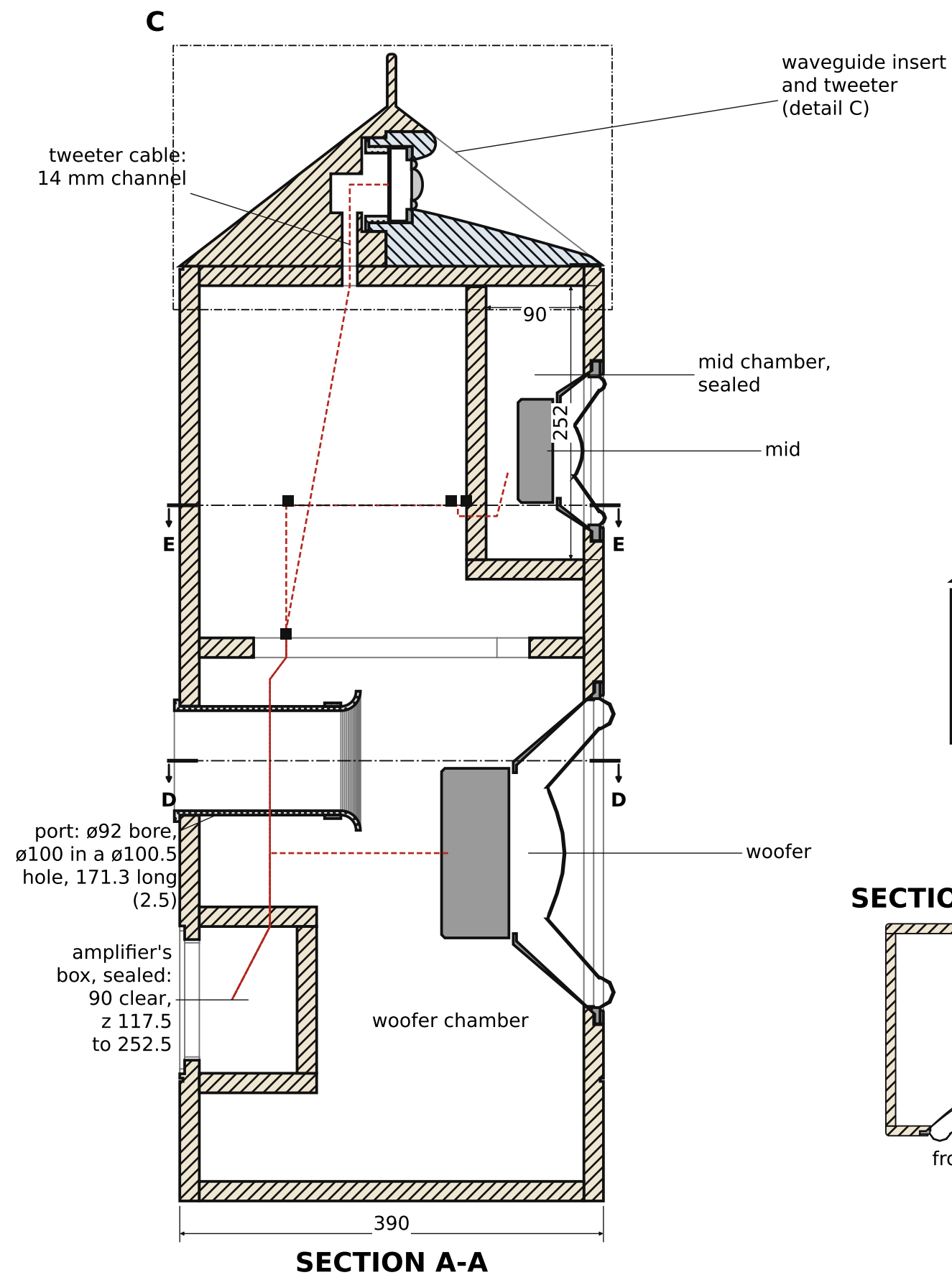


mm | issue 2026-10-09 | HOLD until M2, M3 (the drivers)

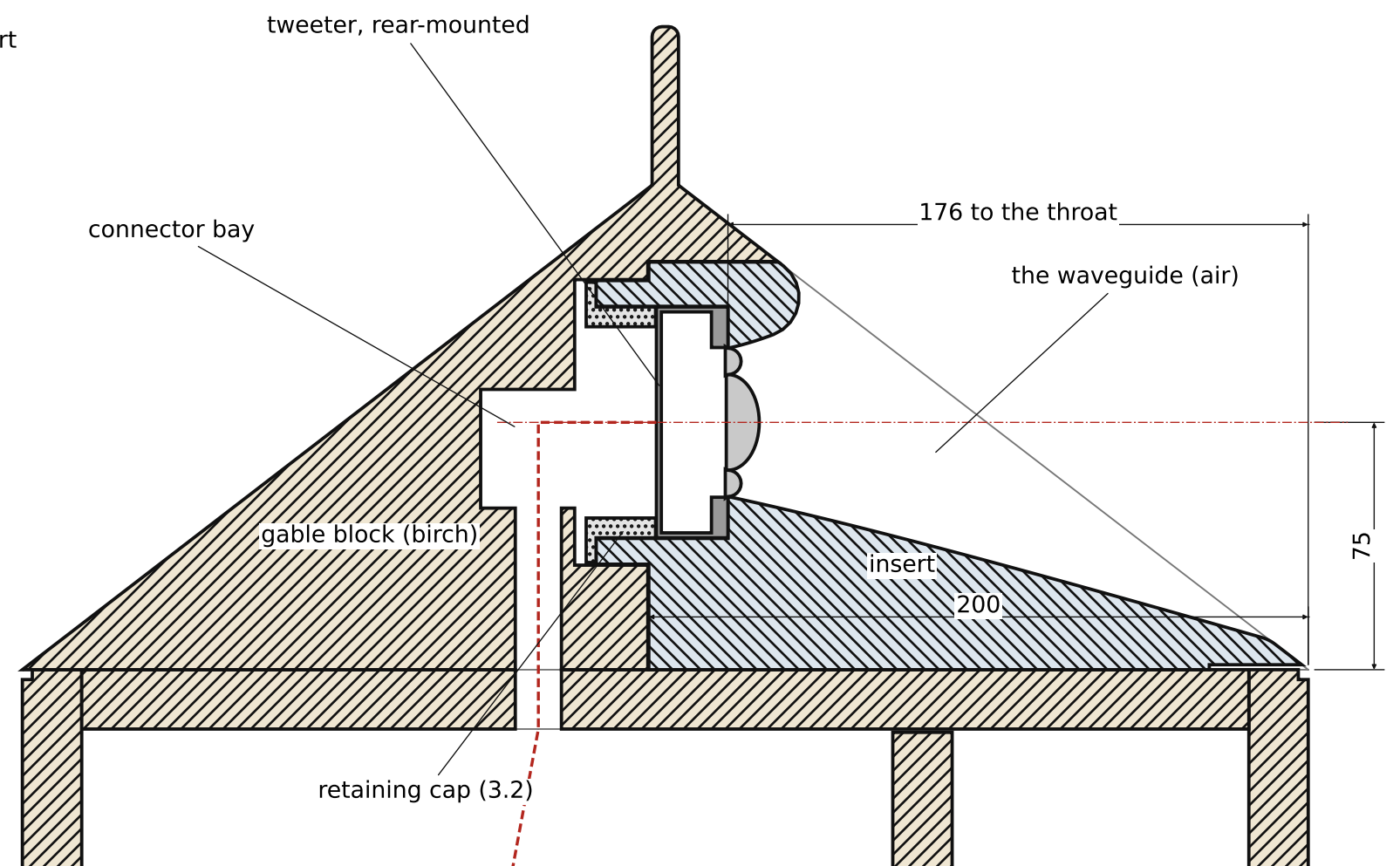
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

1 of 9

EM-FS-001 rev D

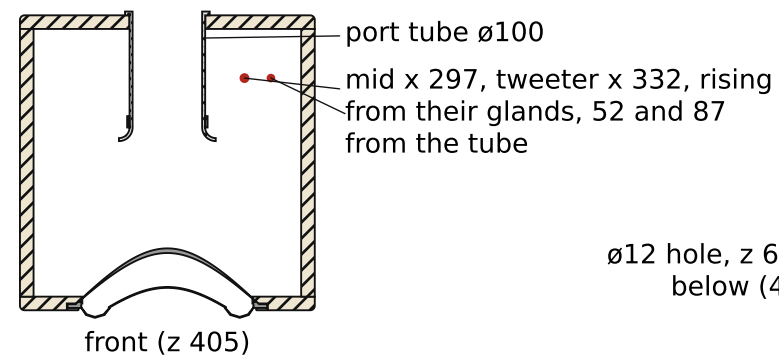


red dashed: the cables' runs to the amplifier (sheet 4),
projected onto the centre plane (in plan: sections D-D and E-E);
black squares: cable-tie mounts (4.6)

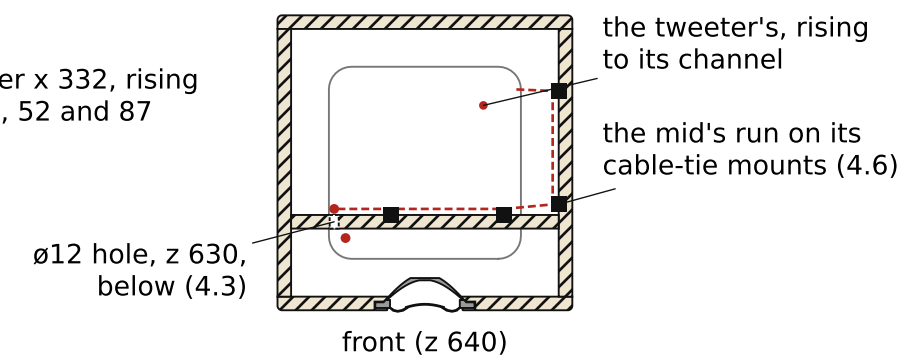


the cable simplified: its connector and the tweeter's own lead on sheet 3
The roof: notes 2.1 to 2.5, sheet 8

SECTION D-D, 1:10



SECTION E-E, 1:10

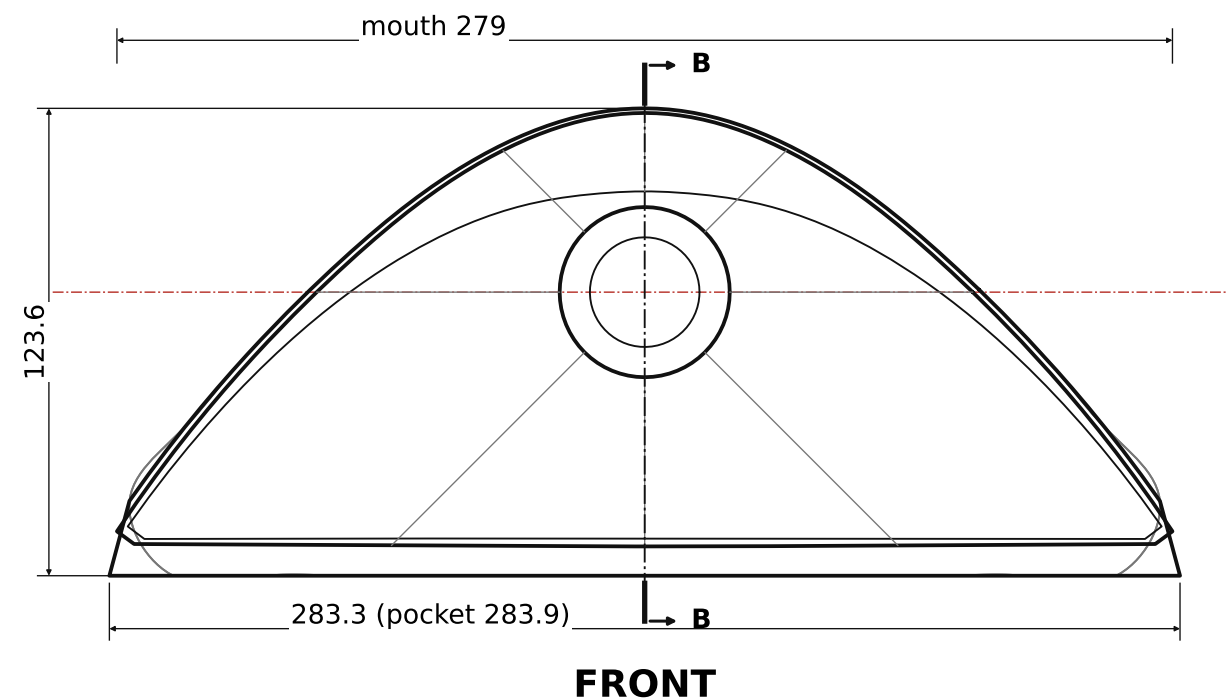


Section on the centreline

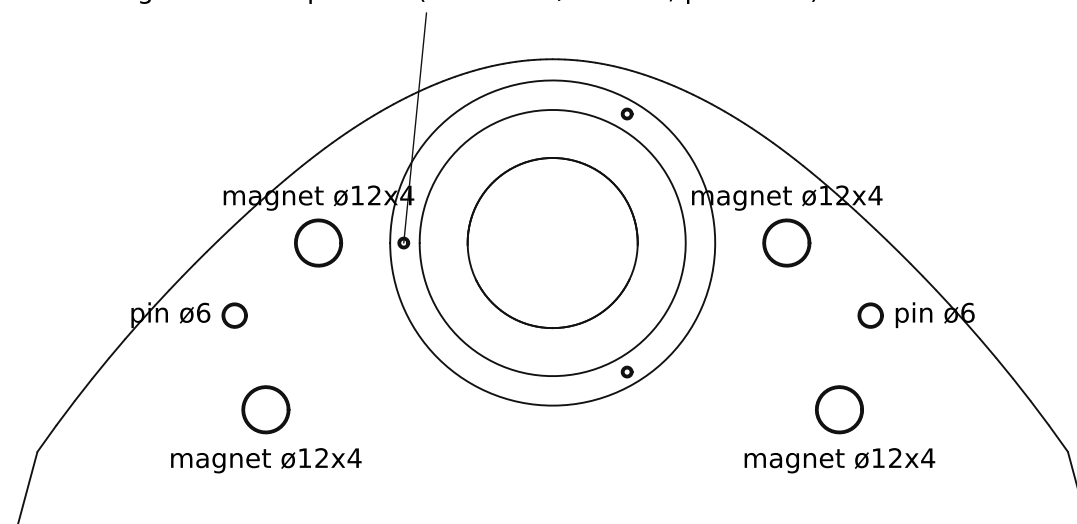
earmilk floorstander. Section A-A on the centre plane (x 195), from the left, at 1:5;
the roof at 1:2. The insert hatched the other way.

mm | issue 2026-10-09 | for reference
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

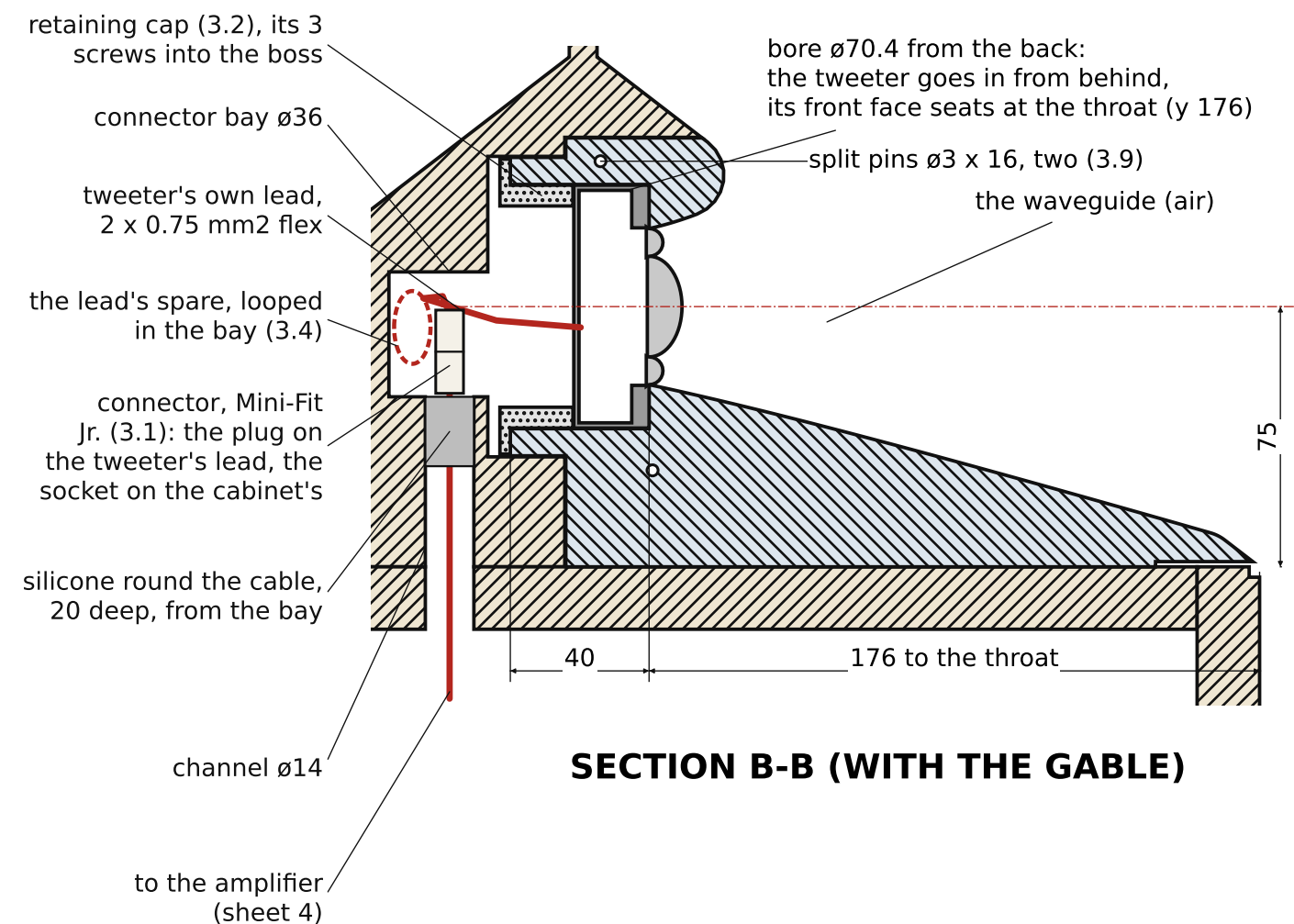
2 of 9
EM-FS-002 rev D



3 x $\varnothing 2.4 \times 10$ pilots on $\varnothing 78.8$ in the boss's back face, one on the horizontal at the speaker's right (left here), 120° apart, for the retaining cap's 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head)

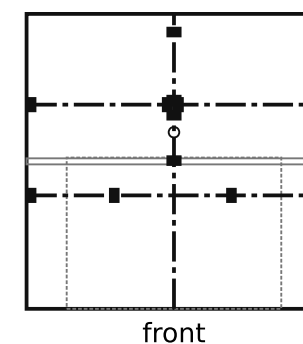


BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert: notes 3.1 to 3.9, sheet 9

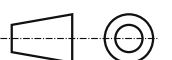
PRINTED GABLE (ROUTE B), PLAN, 1:10



Split planes (chain lines) and pins (bars): stl/gable-print/
 x = 195: (y 196, z 996), (y 256, z 866), (y 276, z 936), (y 366, z 866)
 y = 150: (x 6, z 866), (x 116, z 956), (x 271, z 956), (x 381, z 866)
 y = 270: (x 6, z 866), (x 186, z 936), (x 201, z 866), (x 381, z 936)
 12 steel pins $\varnothing 4 \times 20$ a speaker, in $\varnothing 4.2$ holes 10.5 deep each side; dry-fit each joint closed, then 3M DP420 over its full rim, and seal the channel halves' seam along its length. Hollow the pieces to 3 mm walls, two $\varnothing 3$ drains each in the outer slopes or end faces only (none in the base, a joint face, the channel, the bay or the pocket), plugged with resin and filled before paint.

Waveguide insert and tweeter mount

earmilk floorstander. The insert at 1:2: front, section on the axis, back; the pocket; fitting and service.

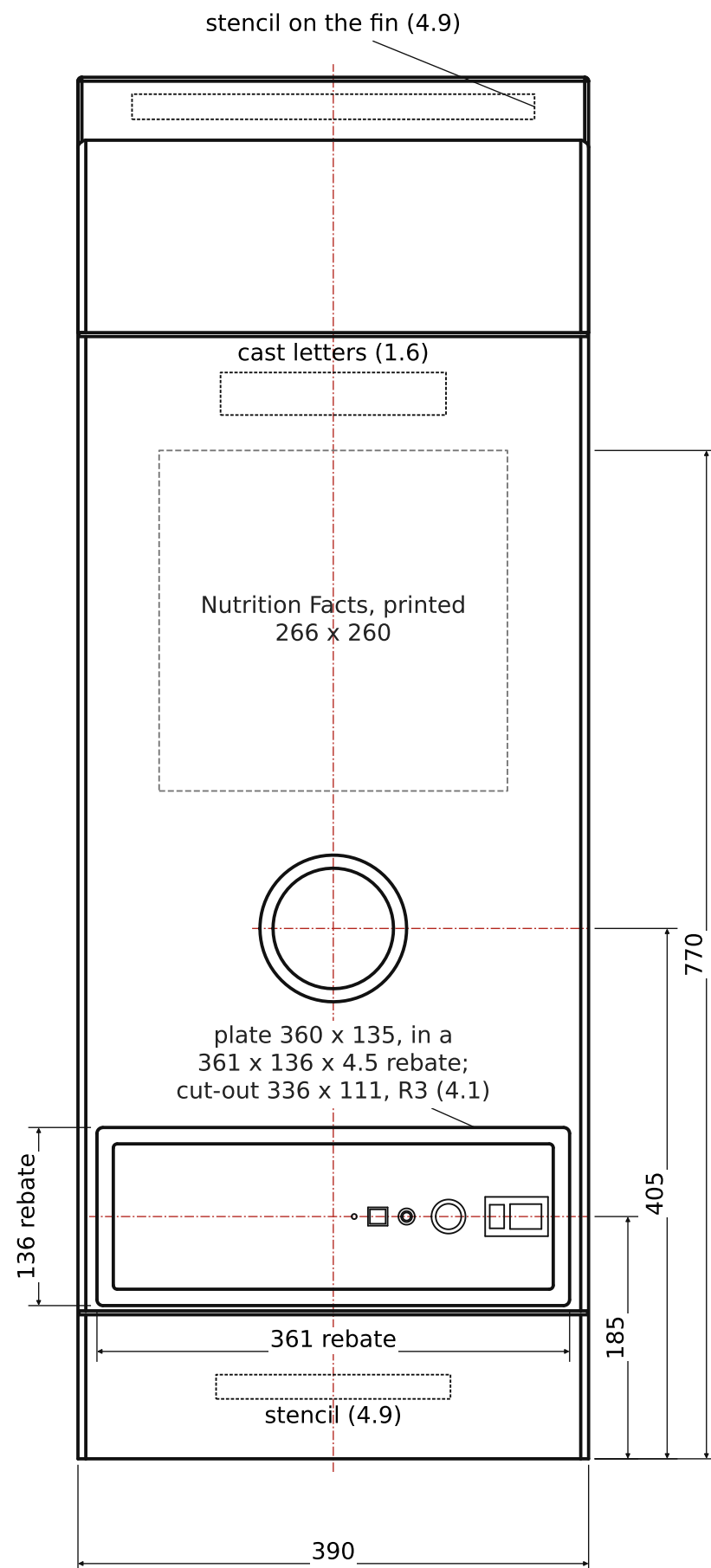


mm | issue 2026-10-09 | HOLD until M4, M6 (tweeter, connector)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

3 of 9

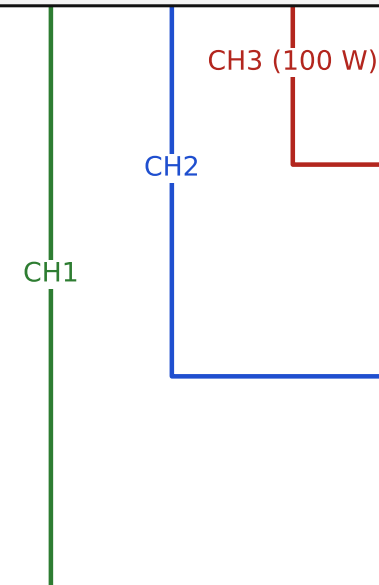
EM-FS-003 rev D



Mains
IEC C14 inlet, switch

Signal in
XLR or RCA
USB for the DSP

Hypex FusionAmp FA253
3 channels, DSP crossover and EQ (ADAU1452)
250 + 250 W into 4 ohm, 100 W tweeter channel
on its side across the back's foot
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)



Tweeter, SB Satori TW29DN-B
29 mm dome, faceplate off, on the insert
1 mm2 pair, red +, black -

Midrange, SB Satori MR16P-8
6.5 in papyrus, its own sealed chamber
1.5 mm2 pair, red +, black -

Woofer, Dayton Audio RSS315HF-4
12 in, 4 ohm, vented to 32 Hz
2.5 mm2 pair, red +, black -

Runs, connectors and seals: notes 4.1 to 4.9, sheet 8

Back and wiring

earmilk floorstander. The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

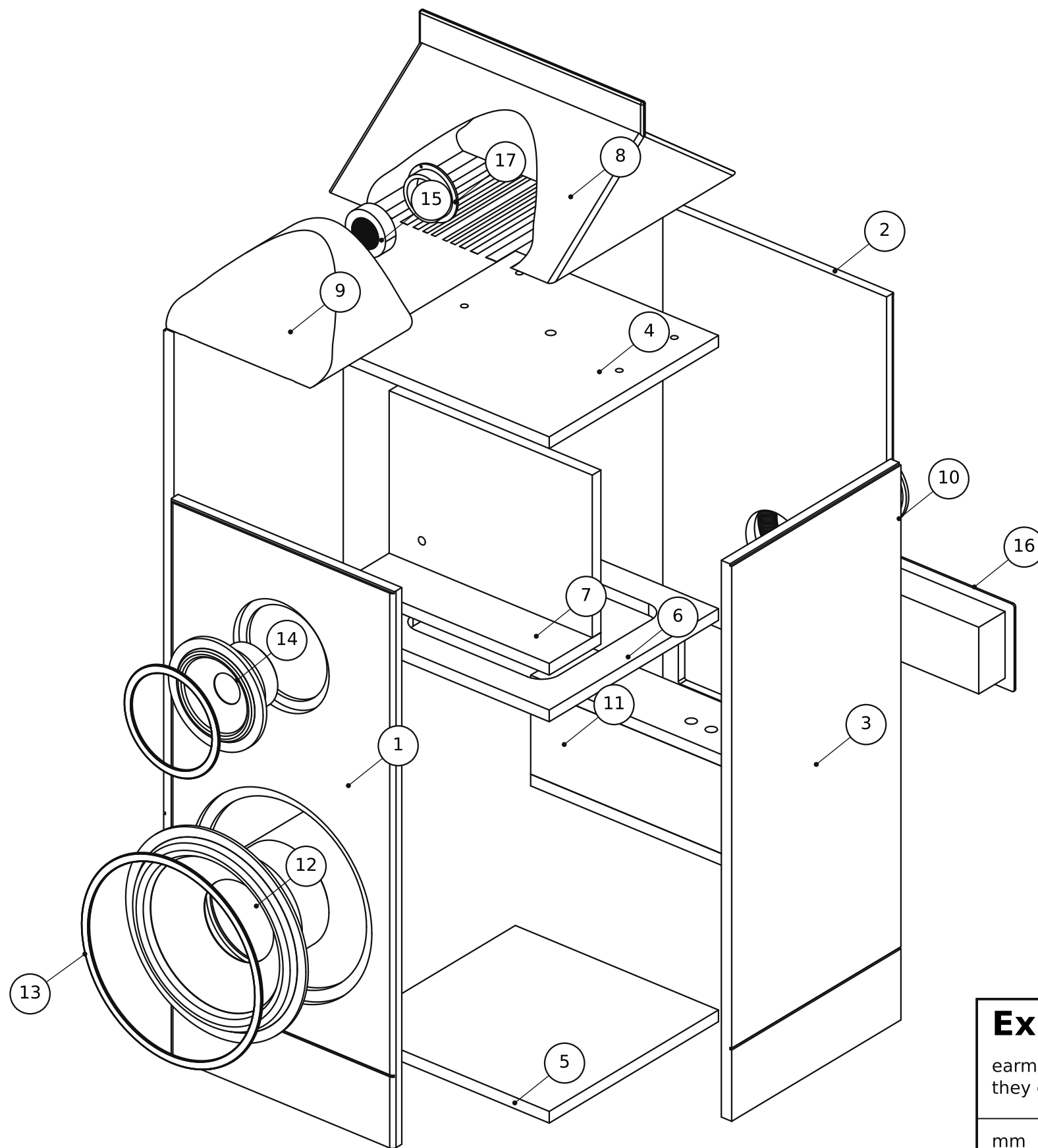


mm | issue 2026-10-09 | HOLD until M5 (the amplifier)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

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EM-FS-004 rev D



Parts

- ① **Front baffle x1**
18 mm Baltic birch, CNC: cut-outs and rebates
- ② **Back x1**
18 mm birch: port hole, amplifier cut-out, rebate
- ③ **Sides x2**
18 mm birch
- ④ **Top panel x1**
18 mm birch: the 14 mm cable hole, 4 dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Window brace x1**
18 mm birch
- ⑦ **Mid shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
- ⑧ **Gable block x1**
laminated birch, CNC, or printed in six pieces (step 1)
- ⑨ **Waveguide insert x1**
SLA tough resin or MJF nylon; magnets, pins
- ⑩ **Port x1**
printed tube and flare collar, 92 bore
- ⑪ **Amplifier box x3**
birch floor, lid (3 glands) and front; sealed
- ⑫ **Woofer x1**
Dayton Audio RSS315HF-4, 12 in, 4 ohm
- ⑬ **Trim rings x2**
printed, satin black, over each frame, screws
- ⑭ **Midrange x1**
SB Acoustics Satori MR16P-8, 6.5 in
- ⑮ **Tweeter x1**
SB Acoustics Satori TW29DN-B, faceplate off
- ⑯ **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels
- ⑰ **Tweeter retainer x1**
printed cap on the motor's back, 3 screws

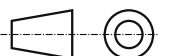
Glue-up order: G1 to G7, sheet 8

Fixings, a speaker (bom.csv)

M4 T-nuts and M4 x 20 low button heads (F1)	12
M16 x 1.5 cable glands and their lock nuts	3
magnets $\varnothing 12 \times 4$ (insert and pocket)	8
pins $\varnothing 6 \times 10$	2
fluted dowels 10×28 (F3)	4
Molex Mini-Fit Jr. 2-circuit plug and socket (3.1)	1 pair
3.0 x 12 thread-forming screws for plastics (cap)	3
steel pins: $\varnothing 3 \times 16$ (insert halves), $\varnothing 4 \times 20$ (printed gable)	2 + 12
foam gaskets 1.5 and 0.5 mm, EPDM tape 3 mm (the plate)	a strip each
4.3 x 16 self-tapping pan heads (the plate)	as drilled

Exploded view and parts

earmilk floorstander. Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

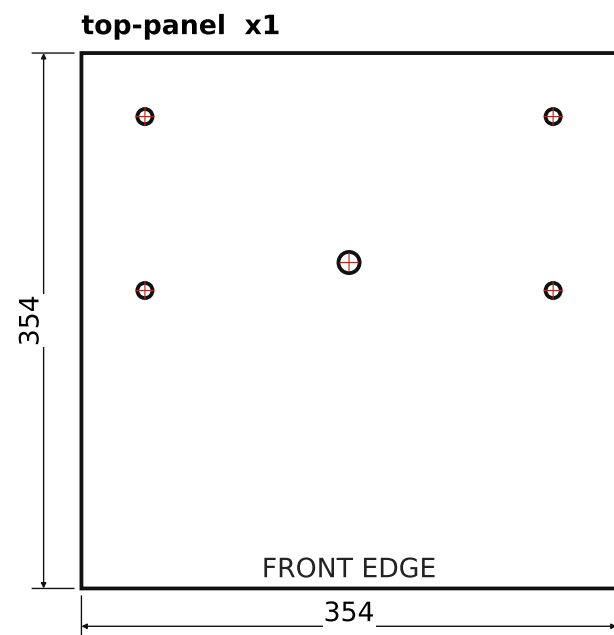


mm | issue 2026-10-09 | for reference

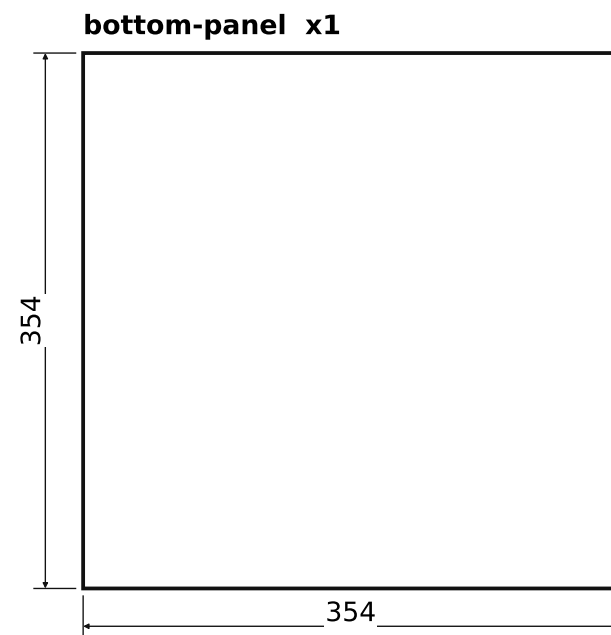
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

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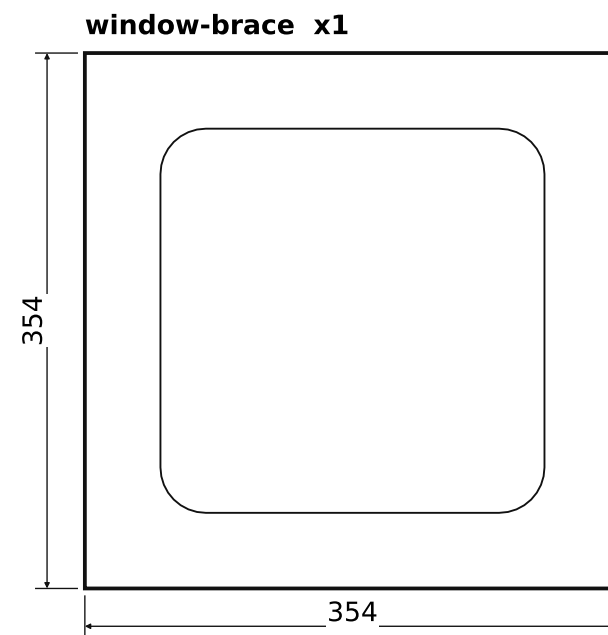
EM-FS-005 rev D



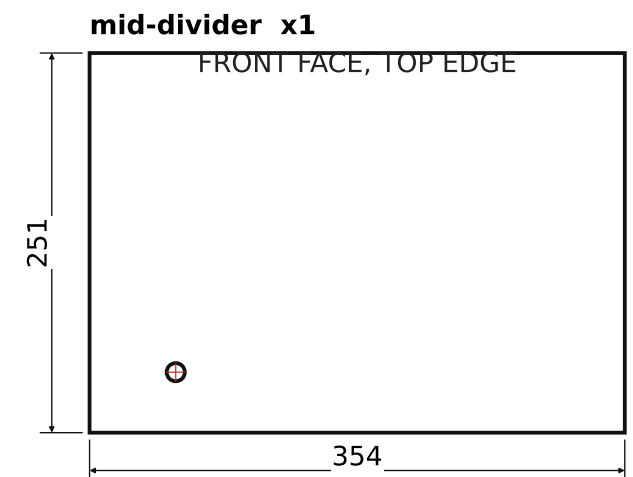
ø14 at (177, 215.5): cut inside
 4 x ø10 at (42, 197), (312, 197), (42, 312),
 (312, 312): drill ø10, 10 deep
 upper face up, FRONT EDGE to the open front
 (the ø14 hole 215.5 from it and 138.5 from the
 back): turned round, the channel opens under
 the insert and the dowels miss the gable;
 dowels register the gable block; the tweeter
 cable's hole is sealed with silicone from the
 bay after wiring



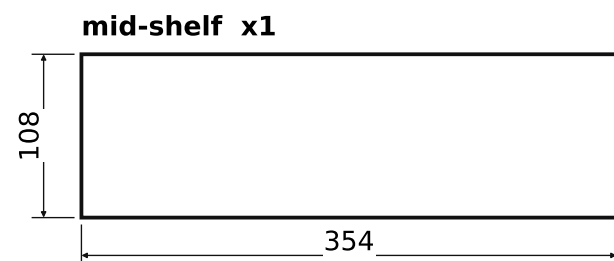
either face



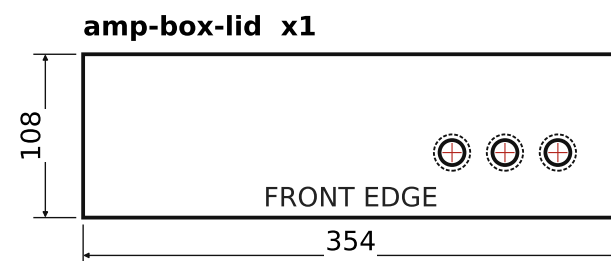
cut inside: 50 to 304 across, 50 to 304 up, R30
 corners
 sits at z 500 to 518, glued to all four walls



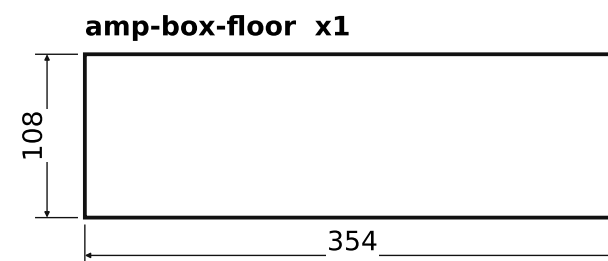
ø12 at (57, 40): cut inside
 seen from the front (the mid chamber's side),
 its top edge up: the hole 57 from its left end;
 y 108 to 126, z 590 to 841, 1 short of the top
 panel (G6's PU fillet closes it); seal the wire
 hole after wiring



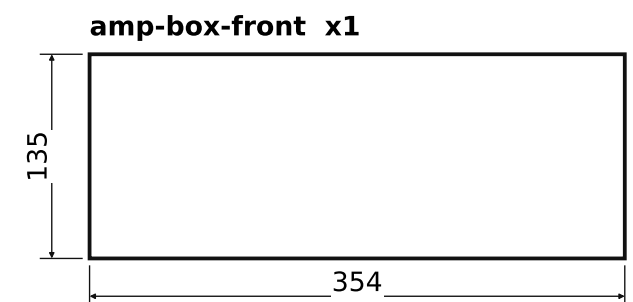
z 572 to 590, from the baffle back to under the divider



3 x ø16.5 at (314, 43), (279, 43), (244, 43):
 cut inside
 3 x ø24 x 8 from the underside (dashed): amp-
 box-lid-underside.dxf, the panel turned over
 left to right, at (40, 43), (75, 43), (110, 43)
 the amplifier box's lid, z 252.5 to 270.5, its
 FRONT EDGE over the box's front; the glands and
 their nuts: note 4.7



the amplifier box's floor, z 99.5 to 117.5, against the back



the amplifier box's front, between floor and lid, 90 in front of the back's inner face; glue and seal all round

Trim rings, section at 2:1 (stl/trim-ring-*.stl): flat, flush with the finish

woofer: ø314 / ø290 x 3



channel open to the bore to ø304.0, 2 deep
 1.5 of each head shows inside the ø290 bore (they
 reach in to ø287; a ø285 bore covers them): owner

the channel clears the M4 x 20 heads (F1): 8 across, 1.6 high or less

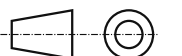
mid: ø165 / ø141 x 3



channel from ø145.0 to ø163.0, 2 deep

Panel details

earmilk floorstander. At 1:5: each panel seen from its outer (upper) face, from its lower-left corner as its DXF; holes from that corner. 18 mm birch.



mm | issue 2026-10-09 | cut after M1; the trim rings HOLD until M2, M3

Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

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EM-FS-006 rev D

Measure first

- M1

Ply: 18 thick; the sides and inner panels are 390 - 2 x 18 = 354 wide. For a real thickness t: 390 - 2t (heights, pockets and rebates unchanged); set WALL = t in params.py and run fab/build.sh again. The gable's layers are t too: 11 for its 195 (one more under t 17.73). The birch under each rebate is t less its depth (woofer t - 9, mid t - 11.5): buy T-nuts whose barrels are 0.5 shorter than that (a CAD check).
- M2

Woofer (Dayton RSS315HF-4): flange 5 thick sets the rebate, 9 deep; the frame's diameter + 1.6 sets the rebate's ø315.6; the cut-out ø282; 8 holes on ø295 (Audiophonics' figure); the surround at its glue line + 2 sets the trim ring's bore, ø290.
- M3

Mid (SB Satori MR16P-8): flange 7.5 thick sets the rebate, 11.5 deep; the frame's diameter + 1.6 sets the rebate's ø166.6; the cut-out ø140.3; 4 holes on ø154; the surround at its glue line + 2 sets the trim ring's bore, ø141.
- M4

SB Acoustics Satori TW29DN-B, its faceplate off, drawn from SB's drawing: one body ø70, 21.7 behind the 5.0 faceplate (29.3 overall). Its widest part behind the dome sets the bore, + 0.4 (ø70.4), which centres it, and the cap's ring, ø70; its depth behind its front face (21.7) sets the cap's length, the boss's back less that (18.05, 0.25 over its gap); its back needs a flat rim 6 wide round its edge, its tabs inside ø58 or on its side. The dome and surround, 45 across, set the throat. Lift the faceplate as SB says, slowly (glue at the diaphragm's rim can stick to it): if the rim is not held without it, the unit needs SB's adapter ring (with the WG29-187) or a printed one on the faceplate's screws, and the bore grows to its outside diameter (README). Print nothing until it is measured.
- M5

Hypex FusionAmp FA253 (Hypex's manual: vertical or horizontal, in its own compartment; here on its side): the plate 360 x 135 sets the rebate, 361 x 136; its corner radius r sets the rebate's corners, R = r + 0.5 (R4.5 for r 4: a square corner, or one under R2.8, will not seat); its thickness t (3) sets the rebate's depth, t + 1.5 (4.5, over the EPDM pressed), and the screws' length, t + 13 (16); the module sets the cut-out, 336 x 111; the screw holes come from the plate in hand.
- M6

Molex Mini-Fit Jr., 2 circuits pair (39-01-2020 and 39-01-2021): its mated length L (24 drawn, from no maker's drawing: Molex SD-5557-003 and 55590020-SD give it) sets the bay, ø36, which leaves 12 over the mated pair for the lead's bend. Print the test bay first (stl/connector-test-bay.stl, README, step 1) and mate and unlatch the pair in it by hand.
- Cut

Cut once M1 is entered: the sides, the bottom and top panels, the window brace, the mid shelf and the divider and the amplifier box's panels (sheet 6). On hold until measured (HOLD in their title blocks and DXFs, stl/HOLD.txt): the front (M2, M3), the back (M5), and every print: the insert, the gable, the cap, the trim rings and the port.

Sheet 1: Notes

- 1.1

Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 6 mm, the fin's edges 3 mm. 3 x 3 mm shadow lines at z 110 and z 857.
- 1.2

The drivers sit flush: each rebate is the measured flange + 1.0 (a 1.5 mm closed-cell foam gasket, pressed) + 3 (the ring): woofer 9 for a 5 flange, mid 11.5 for a 7.5 flange; measure each flange before cutting (M2, M3): the rebate is the flange + 4.0 only while the T-nut's barrel + 0.5 of birch stays under it: 9 under the woofer's (barrel 8 or less), 6.5 under the mid's (barrel 6 or less). The screws and T-nuts as F1, pressed in before the glue-up. A printed trim ring over each flange (stl/trim-ring-*.stl, woofer ø314 / ø290 x 3, mid ø165 / ø141 x 3, flat, a channel in its back over the screws' heads: the sections at 2:1 on sheet 6) held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round. Its inner ø is the surround at its glue line + 2: measure the drivers and print the rings last (the woofer's ring, ø314, needs a bed of 320 or more, or a print service).
- 1.3

The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 935, 176 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
- 1.4

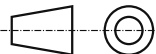
The back (port, amplifier, label) is drawn on sheet 4 with the wiring.
- 1.5

Finish: colour coat and 2K clear; the Facts printed between them.
- 1.6

The cast letters (out/metal/wordmark-letters.*), 1.5 thick and proud: 172.2 x 32.5 of ink, centred across the plinth's front, the badge's centre z 55; on the back the same at z 813. Place and drill them from the 1:1 templates, out/metal/wordmark-template-front.pdf and -back.pdf, after the clear coat.

Notes: measure first; sheet 1

earmilk floorstander. What to measure before cutting or printing, and the notes sheet 1 point to by number.



Sheet 2: The roof

- 2.1
- The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket +0.2/0 and the insert 0/-0.15 on their widths, so 0.30 to 0.48 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.
- 2.2
- The tweeter is rear-mounted: it goes in from behind through a $\varnothing 70.4$ bore and its front face seats on the ring round the throat, its dome and surround filling the 45 mm throat, so the wall runs on from the surround with no step. A printed cap bearing on its motor's back holds it there, screwed to the boss's back face (sheet 3).
- 2.3
- Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 36$ x 35 deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
- 2.4
- Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).
- 2.5
- The port: a 92 bore, its tube 100 outside in a $\varnothing 100.5$ hole through the back, centre z 405; its $\varnothing 112$ x 5 flange glued to the finish. 153.3 from the flange's face, with the 18 flare collar 171.3 long. Printed 163.3 and trimmed at its inner end until the impedance dip sits at the tuning, measured in the sealed box (the glands tightened, the mid fitted, the divider's hole puttied, the fill in, the amplifier's cut-out closed by a scrap board on the EPDM): a DATS V3 (or a sound-card jig) clipped to the woofer cable's free end inside the amplifier's box; trim 2 mm at a time until the minimum between the two peaks is at 32 Hz (README, step 9).

Sheet 4: Runs, connectors and seals

- 4.1
- The amplifier's plate (360 x 135 x 3) lies on 3 mm EPDM tape in its 4.5 deep rebate, R4.5 corners, its module through the 336 x 111 cut-out (R3 corners: a 6 mm cutter or smaller). Drill its screw holes $\varnothing 3.5$ through the 13.5 left under the rebate, from the plate in hand, centred in its flange, 8 or 10 to suit it: 4.3 x 16 self-tapping pan heads (Hypex's 4.3 x 25 kit also works, its tips 7 mm into the sealed box).
- 4.2
- Woofer (CH1): from its gland in the amplifier box's lid straight to the woofer's terminals, 0.6 m of 2.5 mm2. Fit 6.3 mm push-on terminals to suit the driver.
- 4.3
- Mid (CH2): from its gland up through the brace's window, 30 mm in from its back edge, then through the 12 mm hole in the mid chamber's divider (x 75, z 630), 1.0 m of 1.5 mm2. Seal the hole round the cable with putty: the mid's chamber must stay closed.
- 4.4
- Tweeter (CH3): from its gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 1.2 m of 1.0 mm2, ending 77 above the top panel's underside in the socket of the connector (3.1, sheet 3) standing on the bay's floor, fed down from the bay (3.4).
- 4.5
- Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3). The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box. Fill the woofer chamber lightly (about 150 g of polyester fibre), a port's diameter from its inner end and off the amplifier's box.

- 4.6
- Round-sheathed cable (H05VV-F 2 x 2.5, 2 x 1.5 and 2 x 1.0 mm2, about 9.0, 8.0 and 6.4 across), so each gland grips its cable and the silicone seals round the tweeter's. Inside the amplifier's box cut each of Hypex's harness pairs to about 150 mm and butt-splice it to its round cable (woofer 2.5, mid 1.5, tweeter 1.0 mm2; crimped splices under heat-shrink): only round cable passes the glands. The glands, from the right: the tweeter's (x 332), the mid's (x 297), the woofer's (x 262). Hold the runs with adhesive cable-tie mounts (black squares on section A-A and E-E, sheet 2), each stuck on its panel before that panel goes in, a tie threaded loose through it (G1, G3, G4): one on the lid beside each gland; one on the brace's top face beside its window's right edge at (x 334, y 292), for the tweeter's and the mid's together; the mid's on the right side's inner face at (y 290, z 640) and (y 140, z 640) and on the divider's back face at (x 300, z 640) and (x 150, z 640), then down to its hole at (x 75, z 630); the tweeter's rises from the brace's mount straight to its channel at (x 195, y 233.5), held there by the channel's silicone seal (4.5) and pulled taut from below, no mount under the top panel. At step 9 feed each run through its loose ties and pull them tight, reaching in through the woofer's cut-out and the brace's window (the highest, on the side and the divider at z 640, about 180 above the cut-out); keep every run 20 mm clear of the port's flare ($\varnothing 128$ round z 405).
- 4.7
- The amplifier box's lid: 3 M16 x 1.5 long-thread (15 mm) cable glands, one per cable (clamping about 4.5 to 10), in $\varnothing 16.5$ holes 314, 279, 244 from the left side's inner face and 43 from the lid's front edge. Their lock nuts, 20 AF or less (23 across corners) and 5 thick, are epoxied into the $\varnothing 24$ x 8 counterbores under the lid before it goes in (G2), each centred on a gland's body screwed through it and taken out again: a spanner cannot reach a nut sunk 3 deep. After the finish screw the bodies into them from above, feed the cables, then tighten the domes. The box itself is glued and sealed; its front is the woofer chamber's wall.
- 4.8
- Polarity, delay, the crossover (about 2.8 kHz tweeter to mid, from the waveguide study), the woofer's 25 Hz high-pass and a limiter set to its excursion are set in the DSP (out/dsp/): wire red to + throughout and let the filters do the rest.
- 4.9
- The stencils (out/marks/stencil-*.svg), sprayed through vinyl masks before the clear: SHAKE WELL, 26 mm type, centred on the plinth's back at z 55; OPEN OTHER SIDE, 27 mm type with its arrow, centred on the fin's back face.

Glue-up, in order

- G1
- Cut the bottom panel and every inner panel at one fence setting, 354 ± 0.2 wide (tighter than the title block's ± 0.5 : they set and fill the sides' spacing). Lay the back face down. Glue both sides and the bottom to it (the right side with its two cable-tie mounts on its inner face, 4.6); clamp across the sides and check the diagonals.

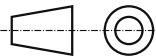
- G2
- The amplifier box: epoxy the glands' lock nuts into the lid's counterbores first (sheet 4); then its floor, front and lid between the sides, against the back, the joints sealed; clamp across the sides.
- G3
- The window brace at z 500, glued to the sides and the back, its cable-tie mount on its top face (4.6); clamp across the sides.
- G4
- Mark the mid chamber on the sides' inner faces: the shelf's underside 554 above the bottom panel, the divider's front face 90 behind the sides' front edges. Glue the shelf, then the divider (its two cable-tie mounts on its back face, 4.6), to the marks; clamp across the sides. The divider is cut 1 short of the top panel (sheet 6), so the top panel slides over it; G6's PU fillet closes the gap.
- G5
- Line the mid chamber with 10 mm wool or polyester felt (spray adhesive): the divider's front face, stopping 15 short of its top edge; the shelf's top; the sides between them; and the top panel's underside over the chamber before it goes in, 10 in from each side and stopping 15 short of the divider. Leave the front's back face bare.
- G6
- The top panel, dowel holes up and its FRONT EDGE to the open front (sheet 6), no cable-tie mounts on its underside; its felt (G5) 334 wide and 75 deep from its front edge, clear of the sides' felt, slid in from the front and glued to the sides and the back, dry over the divider; clamp across the sides. Then, with the front still off, run a 5 mm fillet of PU sealant along the corner between the divider's front face and the top panel's underside, x 18 to 372: the mid's chamber's fourth wall (check it after G7 through the mid's cut-out).
- G7
- Plane any inner panel's front edge that stands proud of the sides flush. Press the T-nuts into the front baffle's inside face; glue the baffle on last, onto the sides, the bottom, the top panel and every inner panel; clamp front to back. When the glue has set, sand the front's top edge flush with the top panel (F2), then fix the gable block (F3).

Fixings and the pocket's floor

- F1
- Drivers: M4 x 20 low-profile (wafer) button head, A2, the head 1.6 high or less and 8 across or less (ISO 7380 is 2.2 high: the trim ring's channel is 2 deep). M4 T-nuts, flange 12 or less across (woofer), 12 or less across (mid) (DIN 1624's M4 is 15: grind each flange flat 6.0 from its centre on the side facing the cut-out), barrel 8 or less (woofer), 6 or less (mid), pressed in before the glue-up.
- F2
- Before the gable block goes on, sand the front panel's top edge flush with the top panel to 0.1: the insert sits on both. Mask the pocket's floor with the insert's sides, base and back when you spray: paint only the insert's face.
- F3
- The gable block on four 10 x 28 fluted dowels (10 deep in the top panel, 20 in the block): a birch block with wood glue; a printed one with epoxy (3M DP420) on its base, scuffed with 120 grit, and on the dowels: wood glue does not hold cured resin.

Notes: sheets 2 and 4 to 6; glue-up

earmilk floorstander. The notes sheets 2 and 4 to 6 point to by number; the order of the glue-up; the fixings.



Sheet 3: Fitting the tweeter and the insert

- 3.1 Solder the tweeter's own lead (320 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each)) to its tabs and crimp the connector's plug on its end: Molex Mini-Fit Jr., 2 circuits, plug housing 39-01-2021 with male terminals 39-00-0041 (18 to 24 AWG) on the tweeter's 0.75 mm2 flex, pin 1 red +. The socket on the cabinet's lead: receptacle housing 39-01-2020 with female terminals 39-00-0077 (16 AWG) on the cabinet's 1.0 mm2 lead. Mated, 24 long or less.
- 3.2 Pass the tweeter in from behind through the ø70.4 bore, dome first, and seat its front face on the throat's seat on a 0.5 mm closed-cell foam ring, ID 46 / OD 70, punched (its ID never inside the throat). Fit the printed retaining cap (stl/tweeter-retainer.stl: a ring ø70 / ø58 x 18.05 long on a flange ø85 x 3) into the bore behind it, the lead through its middle: its ring bears on the motor's back rim and its flange stands 0.25 off the boss. Drive its 3 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head) into the boss until the flange seats, pressing the gasket by that much: no thread to SB's own screw holes is needed, and any motor up to the ring's diameter fits.
- 3.3 Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.
- 3.4 The cabinet's lead is fed down the channel from the bay, through the empty pocket, and caught through the woofer's cut-out (it cannot be pushed up from inside). Its socket stands on the bay's floor, the lead's end 77 mm above the top panel's underside (10 past the floor); seal the channel round it with 20 mm of neutral-cure silicone from the bay before the first fitting. To fit: hold the insert just clear of its pocket, reach in and plug the tweeter's lead into the socket (its 320 mm reaches with a hand in the pocket), then slide the insert home, feeding the spare into the bay as a loop, clear of the 1.1 mm between the cap's screw heads and the bore's floor.
- 3.5 Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0 and the insert 0/-0.15 on their widths (0.30 to 0.48 a side; the boss in its bore the same); paint only the insert's face. A print service holds about 0.2 to 0.3 % (0.6 to 0.9 on 283): order the base's width and the boss's ø86 as critical dimensions, or sand the unpainted sides, base and boss until it slides home with 0.3 to 0.5 a side; never sand the waveguide, the seat or the bore. Check the width before painting.
- 3.6 Service: pull it out by a ribbon loop glued at the back of the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it. Between times the loop folds back into its groove: only its end shows, under the insert's front edge.

- 3.7 The pocket (gable-block.step): the insert's outline + 0.3 a side (283.9 wide at its base), from the slope to its back wall at y 200.3; the boss's bore ø86.6 to y 222.5; the connector bay ø36 from y 216 to 251 on an axis at z 927; the channel ø14 at x 195, y 233.5 down to the top panel; R3 or less in the profile's corners only, and sharp where the back wall meets the walls (a flat-ended cutter), as the insert's back edges are; dowel holes ø10 x 20 in the base at (60, 215), (330, 215), (60, 330), (330, 330). Milled (route A), the back wall is 200.3 and the bay 251 from the front: a cutter with that reach, or print the block (route B).
- 3.8 Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ±75.9 across, 30.9 up; ±61.9 across, 75.0 up; pins ±84.1 across, 55.8 up. Magnets ø12 x 4 in ø12.2 x 4.3 holes, epoxied, poles marked; pins ø6 x 10 bonded with epoxy in ø6.1 x 5.5 in the insert, a slip fit in ø6.2 x 5.5 in the wall (their own tolerances, not the title block's: the insert's pin holes ø6.1 +0.05/0, the wall's ø6.2 +0.1/0, the magnets' ø12.2 +0.1/0 x 4.3 +0.2/0, each depth from its own face).
- 3.9 Printing the insert: one piece from a service (SLA tough resin or MJF nylon, a build of 290 x 220 x 130 or more) is the default, with no seam in the waveguide. On a desktop resin printer (218 x 123 x 220) print it in halves split at the centre plane, the section B-B: each half fits only tilted (about 209 x 117 x 213). Join them with two ø3 x 16 steel dowel pins across the split, at y 175.0, z 887.8 and y 190.0, z 976.9, epoxied in ø3.2 holes 8.5 deep each side (0.5 longer than the pin, for the epoxy); dry-fit the halves closed on the pins before mixing the epoxy; fill and sand the seam on the waveguide's wall flush. MJF prints solid; SLA solid, or hollow to 4 mm walls with two ø3 drain holes in the base face between the pins, plugged with resin: none in the waveguide, the seat or the bore, and none within 5 mm of a magnet, pin, pilot or insert hole.

Notes: sheet 3

earmilk floorstander. The notes sheet 3 points to by number: fitting the tweeter and the insert, printing it.

